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APRIL 20, 1960
SUPERSEDING
Int. Fed. Spec. TT–D–00291a(GSA–FSS)
July 27, 1959 and
Fed. Spec. TT–D–291
December 16, 1947

FEDERAL SPECIFICATION

DIAMYL PHTHALATE: PLASTICIZER (FOR USE IN ORGANIC COATINGS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers diamyl phthalate suitable for use in organic protective coatings.

1.2 Classification.

1.2.1 *Grade and type.*—This specification covers one grade and one type of diamyl phthalate for use as a plasticizer in protective coatings.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP–B–636—Box, Fiberboard.
PPP–C–96—Cans, Metal, 28 Gage and Lighter.
PPP–D–729—Drums, Metal, 55 Gallon (For Shipment of Noncorrosive Material).
PPP–D–760—Drums and Pails, Metal (5 and 16.64 Gallon).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 141—Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the

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procuring activities or as directed by the contracting officer.)

2.2 Other publications.—The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

National Motor Freight Classification Rules and Container Specifications (No. 11 and No. A-1).

(Application for copies should be addressed to the American Trucking Association, Inc., 1424 16th Street, N. W., Washington 6, D. C.)

3. REQUIREMENTS

3.1 Quantitative requirements.—The diamyl phthalate shall meet the requirements specified in table I.

TABLE I.—*Quantitative requirements*

Characteristics	Requirements	
	Minimum	Maximum
Specific gravity at 20°/20°C.	1.022	1.026
Ash, percent by weight	----	0.01
Acidity, mg. of KOH per gram of sample	----	0.067
Refractive index at 20°C.	1.486	1.488
Ester content, percent by weight	99	----
Flash point, °C. (Cleveland Open Cup)	171	----

3.2 Qualitative requirements.

3.2.1 *Appearance.*—The diamyl phthalate shall be clear and free from sediment and suspended matter when examined by transmitted light.

3.2.2 *Color.*—The diamyl phthalate shall be no darker than a solution of 0.0030 gram of reagent potassium dichromate in 1 liter of distilled water.

3.2.3 *Odor.*—The diamyl phthalate shall have the mild characteristic odor of diamyl phthalate.

3.2.4 *Water.* — One volume of diamyl phthalate shall be miscible without turbidity with 19 volumes of 10° heptane at 20°C.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Sampling and Inspection.—Unless otherwise specified, sampling and inspection shall be in accordance with methods 1011, 1021 and 1031 of Federal Test Method Standard No. 141.

4.3 Testing.—Tests shall be made in accordance with the applicable methods of Federal Test Method Standard No. 141 and as specified hereinafter.

4.3.1 The following tests shall be conducted in accordance with Federal Test Method Standard No. 141:

Test	Method No.
Appearance	4261
Color	4247
Odor	4401
Water	4083
Specific gravity	4183
Ash	5262
Acidity	5252
Refractive index	4371
Ester content	5172[1]
Flash point	4294

[1] Factor of 0.153, in lieu of 0.116 in paragraph 3.4 of method 5172.

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5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A.*—Unless otherwise specified, diamyl phthalate shall be packaged in 1-quart or 1-gallon cans conforming to type V, class 4, oblong, of Federal Specification PPP–C–96; in 5-gallon drums conforming to type I, class 1, of Federal Specification PPP–D–760; in 55-gallon drums conforming to type II of Federal Specification PPP–D–729; or shall be shipped in bulk.

5.1.2 *Level C.*—Unless otherwise specified, diamyl phthalate shall be packaged in containers conforming to the Uniform Freight Classification Rules or other regulations applicable to such supplies.

5.2 Packing.

5.2.1 *Level A.*—Unless otherwise specified, twelve 1-quart, or six 1-gallon cans packaged as specified in 5.1.1 shall be packed in a fiber box conforming to type I, compliance symbol W5c, style RSC of Federal Specification PPP–B–636. The 5- and 55-gallon drums require no further packing.

5.2.2 *Level B.*—Unless otherwise specified, diamyl phthalate shall be packed in accordance with 5.2.1 except that fiber boxes shall comply with type I, class 1, style RSC of Federal Specification PPP–B–636.

5.2.3 *Level C.*—Unless otherwise specified, diamyl phthalate shall be packed in containers conforming to the Uniform Freight Classification Rules or other regulations, for safe delivery to destination at the lowest applicable rate.

5.3 Marking.

5.3.1 *Civil agencies.* — Marking for civil agencies shall be in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*—Marking for Military agencies shall be in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use.—The diamyl phthalate covered by this specification is intended for use as a plasticizer in organic protective coating materials. It is a good solvent for nitro-cellulose and is compatible with the usual resins, diluents, and solvents used in the manufacture of lacquers and plastic coatings.

6.2 Diamyl phthalate may be purchased either by volume, the unit being a U. S. gallon at 20°C. (68°F.), or by weight, the unit being 1 pound. The volume of deliveries may be determined by dividing the net weight, in pounds, by the weight per gallon. To obtain the weight per gallon, multiply the specific gravity at 20°/20°C. (68°/68°F.) by 8.33. One gallon of diamyl phthalate at this temperature weighs from 8.513 to 8.547 pounds.

6.3 Ordering data.—Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Applicable levels of packaging and packing (section 5).
(*c*) Quantity per unit package required.

6.4 Transportation description.—Transportation description and minimum weights applicable to this commodity are:

Rail:

Synthetic plastic plasticizer.
Carload minimum weight 36,000 pounds.

Motor:

Plasticizers; paint, lacquer, varnish, gum, resin or plastic.
Truckload minimum weight (W)36.6 pounds, (W)subject to Rule 34, National Motor Freight Classification.

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MILITARY INTERESTS:
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Navy—W
Air Force.

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